

Ensemble

Nimisha Roy
Georgia Tech

Which Classifier/Model to Choose?

Possible strategies:

- Go from simplest model to more complex model until you obtain desired accuracy
- Discover a new model if the existing ones do not work for you
- Combine all (simple) models

Common Strategy: Bagging

(Bootstrap Aggregating)

Originally designed for combining multiple models, to reduce variance and improve classification “stability” [Leo Breiman, 94]

Uses random training datasets (sampled from one dataset)

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Consider the data set $S = \{(X_i, Y_i)\}_{i=1, \dots, n}$

- Pick a sample S^* with **replacement** of size n
(S^* called a "bootstrap sample")

$S \rightarrow$

$$x = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 9 & 10 & 11 & 12 \\ 20 & 21 & 22 & 23 \\ 5 & 6 & 7 & 8 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

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Consider the data set $S = \{(X_i, Y_i)\}_{i=1, \dots, n}$

- Pick a sample S^* with **replacement** of size n
(S^* called a “bootstrap sample”)
- Train on S^* to get a classifier f^*
- Repeat above steps B times to get $f_1, f_2, \dots, f_B$
- Final classifier $f(x) = \text{majority}\{f_b(x)\}_{j=1, \dots, B}$

Common Strategy: Bagging

Why would bagging work?

- Combining multiple classifiers reduces the variance of the final classifier

When would this be useful?

- We have a classifier with high variance

Bagging decision trees

Consider the data set S

- Pick a sample S^* with replacement of size n
- Grow a decision tree T_b
- Repeat B times to get $T_1, \dots, T_B$
- The final classifier will be

$$f(x) = \text{majority}\{f_{T_b}(x)\}_{b=1, \dots, B}$$

Random Forests

Almost identical to bagging decision trees,
except we introduce some randomness:

- Randomly pick m of the d available attributes, at every split when growing the tree (i.e., $d - m$ attributes ignored)

Bagged random decision trees
= **Random forests**

Random Forests: Why does it work?

1. Reduces variance through bootstrapping

Each tree is trained on a different random sample → averaging cancels noise and stabilizes predictions.

2. Computationally cheaper than growing full trees

Decision trees can be expensive with many features;

Random Forests speed this up by splitting on **only a small subset of features** at each node.

3. Reduces correlation across trees

Randomly selecting a subset of features at each split forces trees to grow differently → diversity leads to a stronger ensemble.

4. Combines many weak learners into one strong model

Individual trees may overfit, but averaging many overfitted trees produces a smooth, robust decision boundary.

5. Automatically captures complex patterns

Learns nonlinear relationships and high-order interactions without feature engineering.

6. Works well out-of-the-box

Handles noise, missing values, outliers, mixed data types, and requires little tuning.

What are our Hyper-Parameters in Random Forest

m = Number of randomly chosen attributes

Usual values for $m = \sqrt{d}, 1, 10$

d is number of dimensions or features or attributes

How to optimize m ? Cross-Validation

B = Number of models or decision trees in Random Forest

Keep adding trees until training error stabilizes (reaches to a plateau)

Important points about random forests

Algorithm (hyper) parameters

- Size/#nodes of each tree
 - as in when building a decision tree
- May randomly pick an attribute, and may even randomly pick the split point!
 - Significantly simplifies implementation and increases training speed
- PERT - Perfect Random Tree Ensembles
<http://www.interfacesymposia.org/I01/I2001Proceedings/ACutler/ACutler.pdf>
- Extremely randomized trees
<http://orbi.ulg.be/bitstream/2268/9357/1/geurts-mlj-advance.pdf>